



EDITORIAL NOTE

Bad Science – what is in the pipeline?

ALEXANDER W.A. KELLNER

As has been pointed out several times, the challenges for scientific editors are increasing at a pace that has been difficult to follow. The reasons are diverse, including new procedures and technologies, such as the advent of Artificial Intelligence (AI), which is becoming increasingly more accessible to the public, including scientists, causing editors nightmares about the validity of submitted manuscripts and how to detect problematic studies (e.g., Kellner 2023a). Not to mention the diverse approaches concerning the evaluation of the quality of papers, a controversy in itself (e.g., Amaral 2024).

One issue, however, is more and more entering the radar of scientific editors and heading towards becoming a major problem: bad science! Yes, poor scientific papers have been around for as long as science has existed, but the facility of information distribution has been key for bad science potentially influencing people in their decisions.

Errors in studies can happen even though we all try to avoid them—authors, editors, and reviewers. The fact is that preparing a paper to present the results of a study that has sometimes been ongoing for years is not easy. Especially for those academics who are not native speakers of English, the *lingua franca* of the world, including science. The only researchers who have never made a simple mistake in their papers—omitting a word, using an awkward expression, overlooking unwanted editorial changes introduced by third parties during the review phase, or even forgetting to include a reference to a publication cited in the text (far from any type of plagiarism)—are those who are not very productive. These types of so-called honest mistakes are not intentional and do not compromise the results of the study. In most cases, they can be easily rectified with corrections and the publication of an erratum. No harm, no hard feelings.

There are also other cases where there is debate between researchers with different views on a given topic (e.g., Coimbra 2023, Leite et al. 2023). The editors provide both sides with opportunities to debate the issue and the scientific community can judge for itself who is right. This is how science works (or should work). No stress.

On the other hand, there have been several instances where scientific malpractice and misconduct are clear and intentional. This has become even more evident during the COVID-19 pandemic, which has caused millions of deaths and, in some cases, has been the target not only of deliberate misinformation about ineffective treatments (e.g., Sachs et al. 2022), but also of bad science (e.g., Saha 2023). Not to mention the emotional stress on the population that is still being assessed (e.g.,

Schuch et al. 2023). The problem was so severe that there are even some specific studies dedicated to retracted articles involving the latest pandemic (e.g., Cortegiani et al. 2021).

Just to give a brief example of how the issue of bad science is increasing, before 2009, there were less than 1000 yearly retractions, but in 2023, the figure rose to almost 14,000 (Cabanac 2024). Yes, if one takes into account all published papers, this number is still small, but the general feeling is that the problem is increasing. Perhaps the larger number of retractions shows that editors and researchers are more aware of the problem. On the other hand, authors might have seen more advantages than drawbacks in this unethical behavior.

Why are scientists driven to produce bad science? Perhaps the easiest answer is because they are humans. This question can also be directed at why there are still wars, when everyone knows the negative results of *Homo sapiens* in settling their disputes by violence. Bad behavior is observed in any profession and scientists are no exception. Another reason has to do with the current scientific environment of what is known as “publish or perish”, a topic that can lead to quite complex analyses (e.g., Amutuhaire 2022). But the fact is that the academic system is based on what has also been called the “bakery effect” - the need for “fresh publications” at pre-established times (see Kellner & Ponciano 2008). Like it or not, every scientist must publish in order to get a job and, above all, keep it! There are cultural differences in the way researchers are evaluated and deal with pressures regarding their careers, but the drive to publish more and more is present and does not seem to be decreasing in the near future. Temptations to take shortcuts, including a variety of illicit practices (some as extreme as buying authorships in third-party publications), exist and appear to outweigh the risks (see Bolland et al. 2025). Even after retraction, bad papers still receive a lot of attention on social media and other websites, suggesting that retraction is less effective than one might imagine, given the online attention that studies with misleading results receive (e.g., Peng et al. 2022). Somehow, this malpractice seems to be either ignored or downplayed without major negative consequences for those who practice it.

The problem is here, and, apparently, we do not really know how to deal with it. Generalized and widespread sanctions and punishments are hard to enforce, for all considering that most journals depend on the goodwill of authors, editors, and reviewers that work *pro bono* to the sake of the system. One way out might be to use the media - including social media. Studies have shown that once authors are exposed for bad science, they lose credibility and are less likely to publish new studies (Alabrese 2022). Efforts to enforce more rigorous peer review should be encouraged, but there is a general feeling that the peer review system cannot be considered by itself an effective tool for detecting problematic publications (e.g., Ramasamy 2021). Not to mention the difficulties in obtaining reviewers today, a major problem for the publishing industry (e.g., Kellner 2023b).

The issue of bad science is on the rise and sooner or later it will need to be addressed seriously. Even though the number of cases is still very small compared to the total number of published articles, the problem affects the credibility of science and has been used selectively against scientists, and may soon affect research funding in general. Perhaps new technological tools could be useful to help editors keep everyone in the publishing industry honest. Not an easy task.

REFERENCES

- ALABRESE E. 2022. Bad Science: Retractions and Media Coverage. CESifo Working Papers 10195: <https://ssrn.com/abstract=4324218>.
- AMARAL OB. 2024. Academic impact and scientific quality should not be conflated. *An Acad Bras Cienc* 96: e20240738. DOI 10.1590/0001-3765202420240738.
- AMUTUHAIRE T. 2022. The Reality of the 'Publish or Perish' Concept, Perspectives from the Global South. *Pub Res Q* 38: 281-294. DOI 10.1007/s12109-022-09879-0.
- BOLLAND MJ, AVENELL A & GREY A. 2025. Publication integrity: what is it, why does it matter, how it is safeguarded and how could we do better? *J Roy Soc New Zealand* 55: 267-286. DOI 10.1080/03036758.2024.2325004.
- CABANAC G. 2024. Stop bad science propagating through the literature. *Nature* 632: 977-979.
- COIMBRA JC. 2023. Rebuttal of Leite and Do Carmo 2021 'Description of the stratotype section and proposal of hypostratotype section of the Lower Cretaceous Quiricó formation, São Francisco Basin, Brazil'. *An Acad Bras Cienc* 95: e20220133. DOI 10.1590/0001-3765202320220133.
- CORTEGIANI A, CATALISANO G, IPPOLITO M, GIARRATANO A, ABSALOM AR & EINAV S. 2021. Retracted papers on SARS-CoV-2 and COVID-19. *Br J Anaesth* 126(4): e155ee156. DOI: 10.1016/j.bja.2021.01.008.
- KELLNER AWA. 2023a. Artificial Intelligence in scientific publications? *An Acad Bras Cienc* 95: e202395S1. DOI 10.1590/0001-37652023202395S1.
- KELLNER AWA. 2023b. Looking for reviewers - where did they go? *An Acad Bras Cienc* 95: e202395S2. DOI 10.1590/0001-37652023202395S2.
- KELLNER AWA & PONCIANO LCMO. 2008. H-index in the Brazilian Academy of Sciences - comments and concerns. *An Acad Bras Cienc* 80: 771-781. DOI 10.1590/S0001-37652008000400016.
- LEITE AM & DO CARMO DA. 2023. Response to Coimbra (2023) 'Rebuttal of Leite and Do Carmo, 2021'. *An Acad Bras Cienc* 95: e20230200. DOI 10.1590/0001-3765202320230200.
- PENG H, ROMERO DM & HORVÁT E-A. 2022. Dynamics of crossplatform attention to retracted papers. *Proc Natl Acad Sci* 119(25). <https://pnas.org/doi/full/10.1073/pnas.2119086119>.
- RAMASAMY A. 2021. Is Peer Review a Hindrance to Good Science or a Gatekeeper Against Bad Science? *J Oral Maxillofac Surg* 79: 745-747. DOI 10.1016/j.joms.2020.11.033.
- SACHS JD ET AL. 2022. The Lancet Commission on lessons for the future from the COVID-19 pandemic. *Lancet* 400: 1224-1280.
- SAHA K. 2023. Underreporting of COVID-19 Deaths in India: Bad Science or Smart Politics? *Med Res Arch* 11(2): DOI 10.18103/mra.v11i2.3497.
- SCHUCH HS, CADEMARTORI MG, DIAS VD, LEVANDOWSKI ML, MUNHOZ TN, HALLAL PC & DEMARCO FF. 2023. Depression and anxiety among the University community during the Covid-19 pandemic: a study in Southern Brazil. *An Acad Bras Cienc* 95: e20220100. DOI 10.1590/0001-3765202320220100.

How to cite

KELLNER AWA. 2025. Bad Science – what is in the pipeline? *An Acad Bras Cienc* 97: e20259702. DOI 10.1590/0001-376520259702.

ALEXANDER W.A. KELLNER

<https://orcid.org/0000-0001-7174-9447>

Universidade Federal do Rio de Janeiro, Museu Nacional, Laboratório de Sistemática e Tafonomia de Vertebrados Fósseis, Departamento de Geologia e Paleontologia, Quinta da Boa Vista, s/n, São Cristóvão, 20940-040 Rio de Janeiro, RJ, Brazil

E-mail: kellner@mn.ufrj.br

